Angler's Riverfly Monitoring Initiative Scores

Hello Birmingham River Champions. Although we took a detailed look at the ARMI score trigger levels in our July newsletter, we wanted to take another deep dive since there have been more samples rolling in and a little more interpretation. We will also be sending this to the Environment Agency who set some 'trigger levels' at some sites with longer datasets to see whether any need revising and if any new ones can be set with more samples rolling in.

What is an ARMI score?

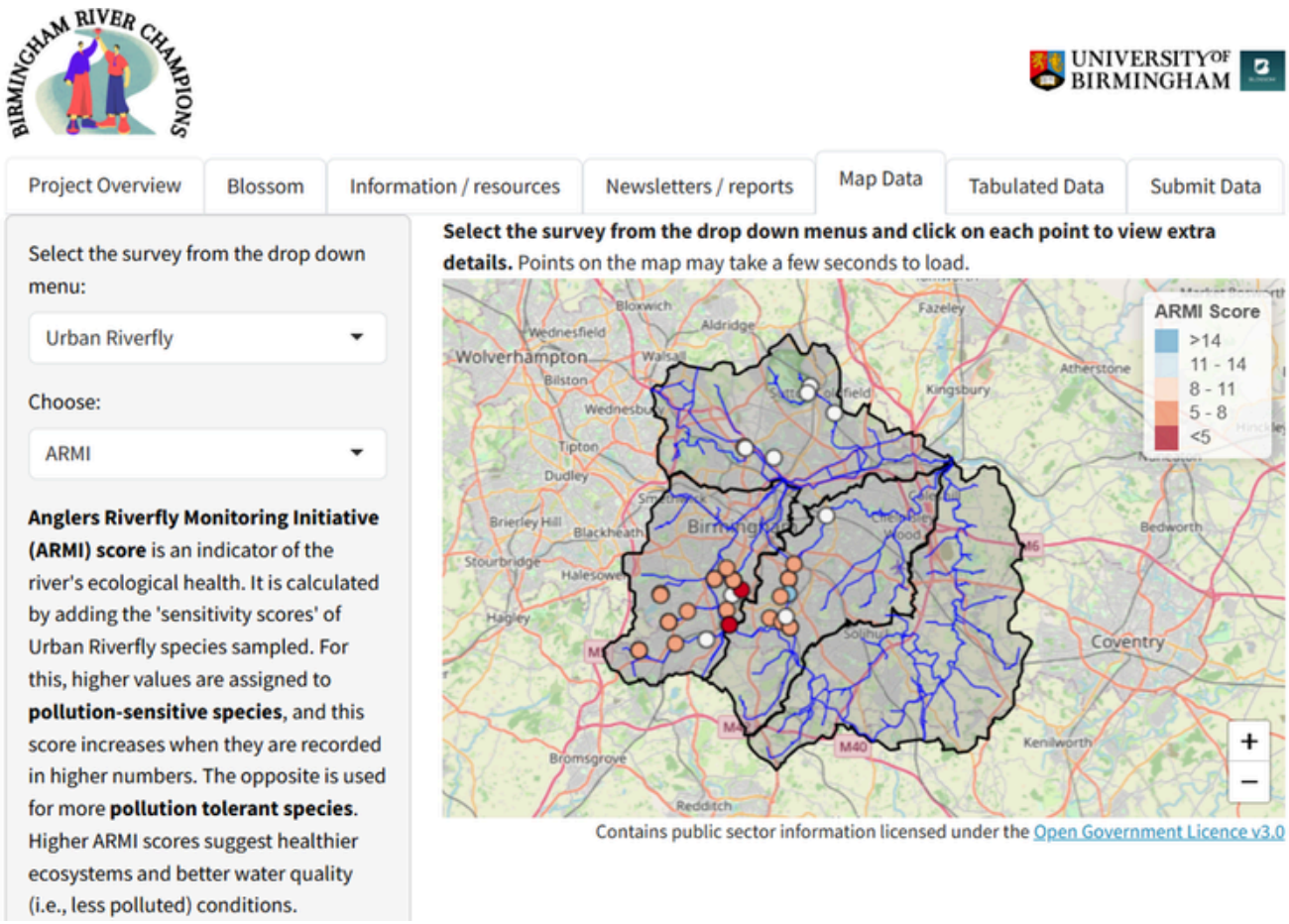
Anglers Riverfly Monitoring Initiative (ARMI) score is an indicator of the river's ecological health. It is calculated by adding the 'sensitivity scores' of Urban Riverfly species sampled.

For this, higher values are assigned to pollution-sensitive species, and this score increases when they are recorded in higher numbers. The opposite is used for more pollution tolerant species. Higher ARMI scores suggest healthier ecosystems and better water quality (i.e., less polluted) conditions.

But there are things to bear in mind (these things are never straight forward)! ARMI can depend on how effectively species are being recorded - such as a size 10 welly collecting more bugs than a size 5 one! Also, the ARMI score has not been widely tested to Urban Riverfly, so Birmingham River Champion's (BRC) data will help improve the reliability of this national recording scheme (see here for further details).

Average ARMI Score

First, taking a look at averaged Angler's Riverfly Monitoring Initiative (ARMI) scores across the city, there is a 'sea of red' – you can take a lot at this any time by going to our web page and clicking on mapped data (screenshot below).

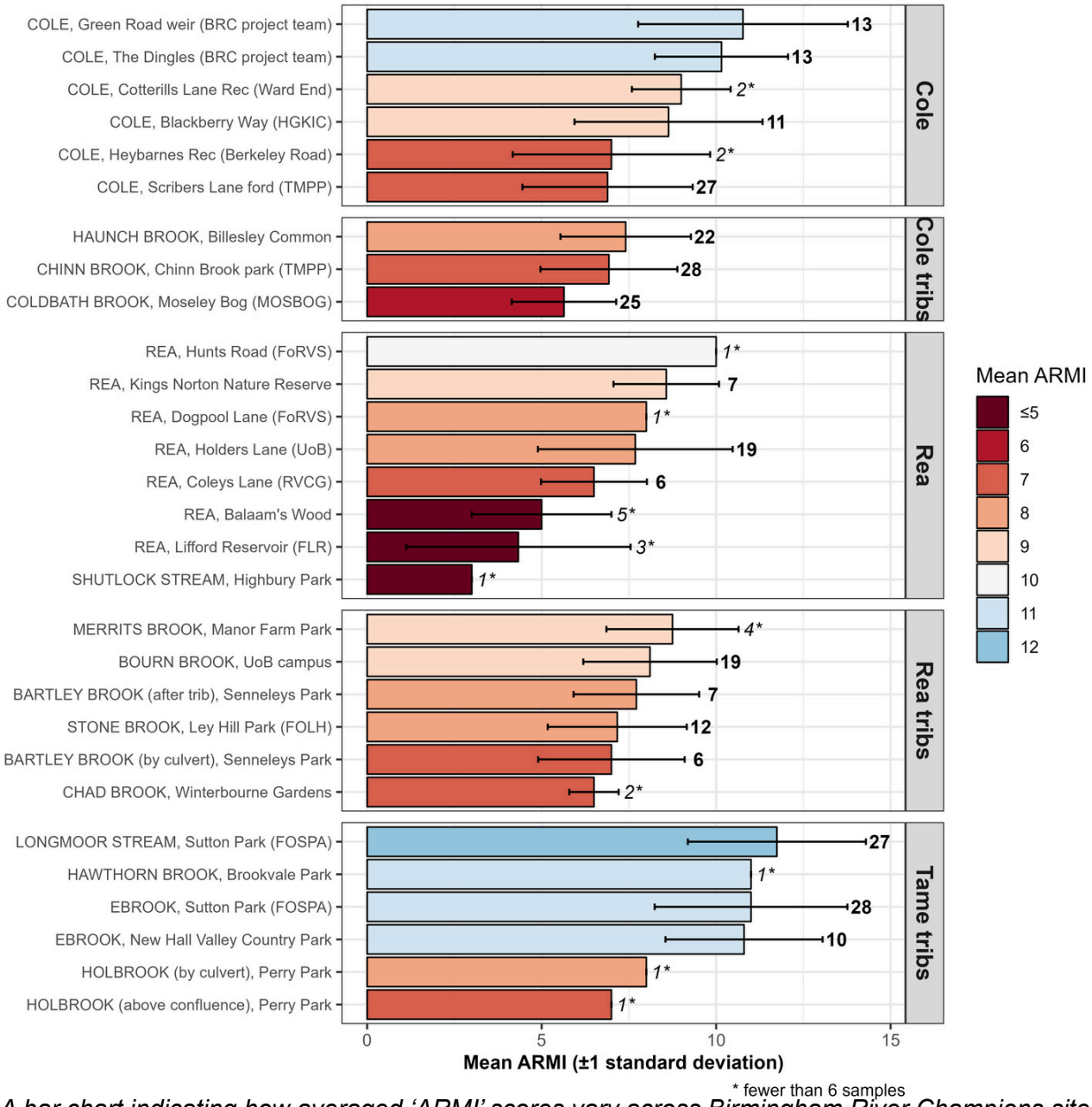


Screenshot of the Birmingham River Champions web application displaying averaged 'ARMI' scores

You can also see this in the bar chart below. For many sites with lower ARMI scores this likely due to the highly urbanised environment of Birmingham's rivers and the various pressures that come with this (pollution, simplified habitats, and flashy flows being a few).

More positive signs arise in some of the Tame tributaries and particularly Sutton Park, where sensitive species like green drake mayflies and flat-bodied stone clinger stoneflies have been found!

The higher ARMI scores along parts of the Cole likely reflect differences in sampling teams (see below – as we also discussed in this report). Two other things to take away from the bar chart below: first, if you look at the error bars, this indicates that ARMI scores can vary quite considerably around its average; second, the number of samples varies quite considerably between sites, and more samples means we'll have greater confidence. Both highlight the importance of repeat monitoring.



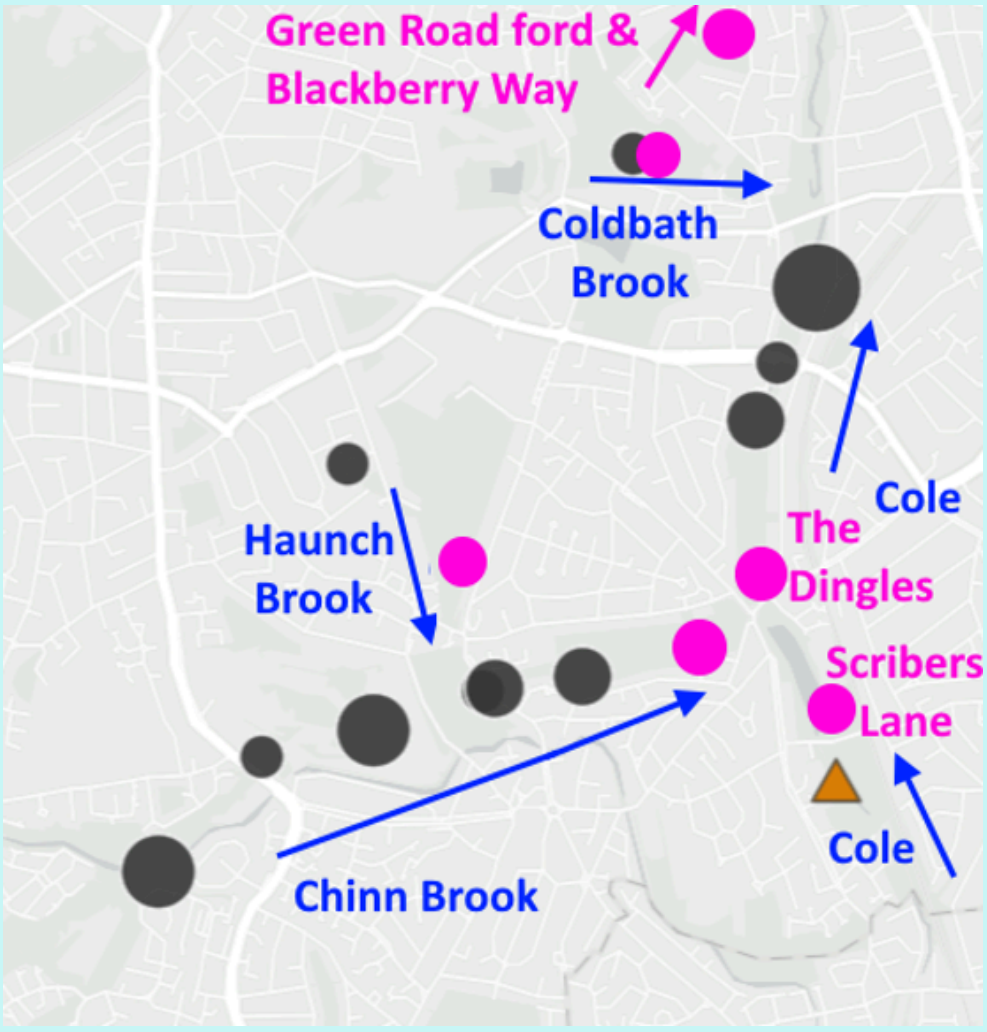
A bar chart indicating how averaged 'ARMI' scores vary across Birmingham River Champions sites

Changes in Time: The Cole

Moving from differences between sites to changes through time, most of the sites along the Cole that have been more frequently sampled are located in its upper reaches and tributaries around the Hall Green area.

Of these, only the Scribers Lane and The Dingles sites on the mainstem are not subjected to upstream Combined Sewer Overflow (CSO) sewage discharges, but are likely subjected to other pressures such as both agricultural and urban runoff - and potentially sewage inputs in the form of misconnections in 'surface water' sewers (we are not aware of any evidence of this currently though).

The Chinn Brook is subjected to more intense CSO inputs, which in turn feed in to the Green Road ford and Blackberry Way sites downstream in the Cole – this latter site is also subjected to severe sewage inputs from misconnections. You can see information on live and historic CSO spills via respective links.

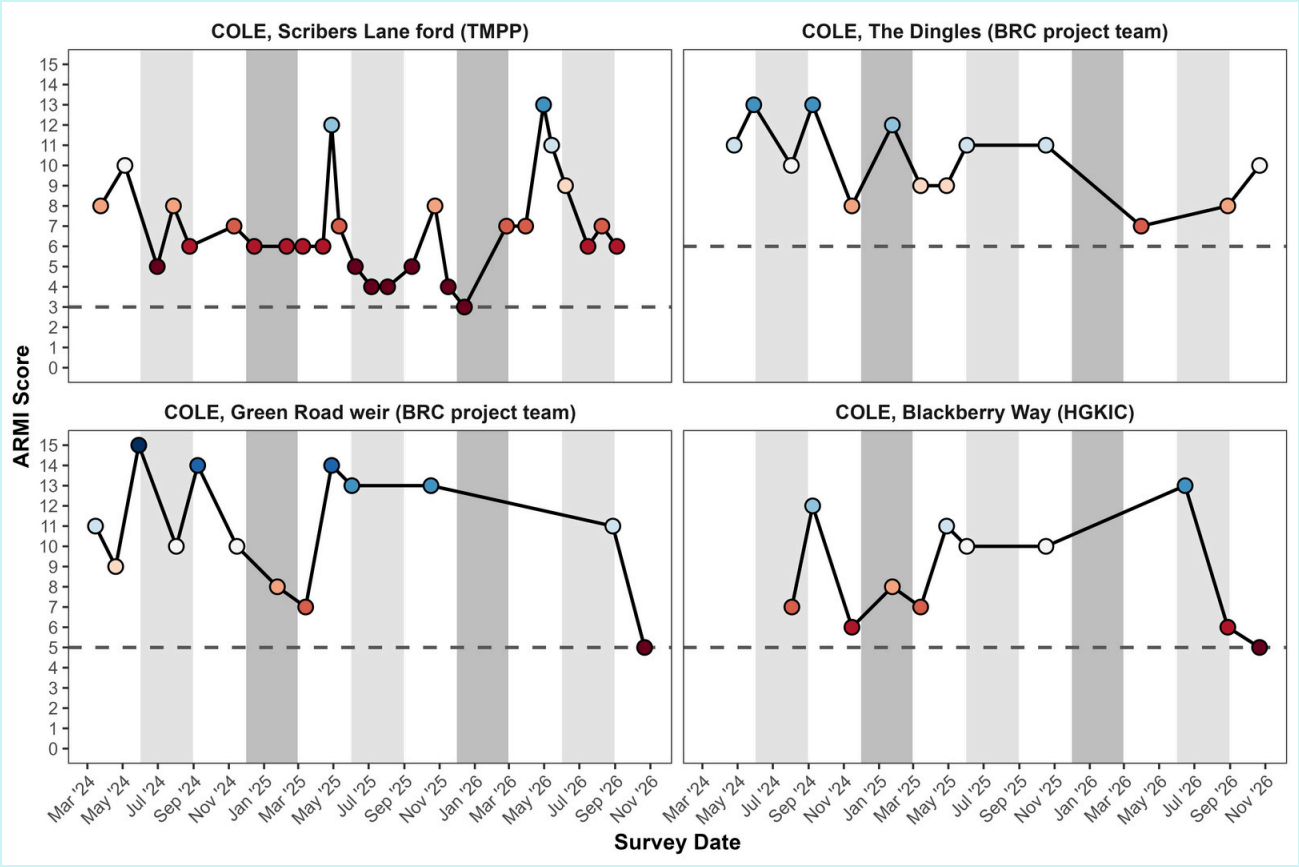


A annotated screenshot from the Environment Agency's 'Event Duration Monitoring - Storm Overflow Explorer' (see link above). The black circles indicate CSOs, with larger sizes reflecting longer periods of time sewage is being released. The pink circles denote Birmingham River Champions sampling sites.

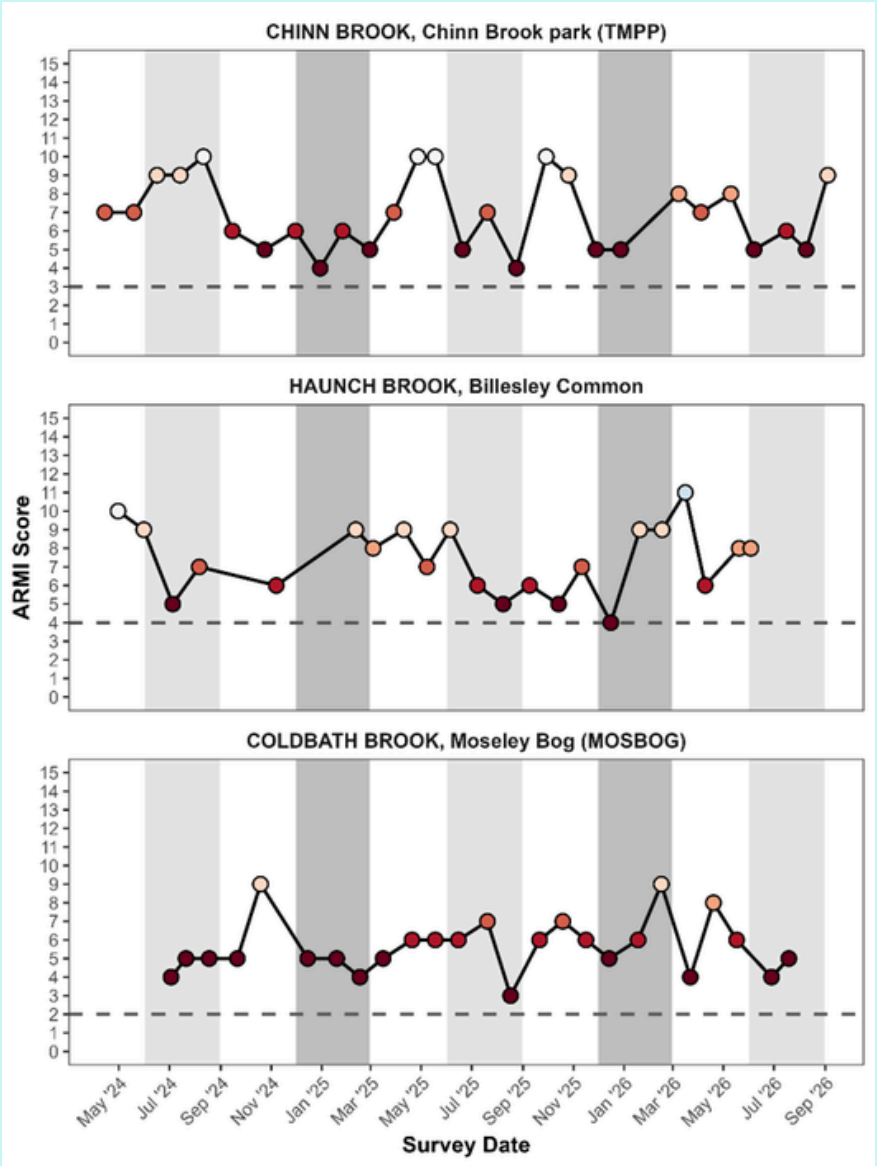
Sites along the Cole shows considerable variation in ARMI scores, although this was slightly less pronounced in its tributaries – and at the Coldbath Brook in particular that has displayed consistently low values. ARMI scores at the Cole at Scribers Lane and Haunch Brook declined to the trigger levels for a single sample last winter, but have bounced back since.

Recently, ARMI scores along the Cole at Blackberry Way and Green Road weir dropped to the trigger level threshold, so we shall return soon to see whether scores go back up - the Environment Agency can investigate if trigger levels are breached twice consecutively

However, as discussed in a recent report of ours, ARMI may not always fully reflect the presence of particularly sensitive species. For example, the pollution-sensitive stone clinger (flat-bodied mayfly) has been recorded at Scribers Lane and The Dingles, but has not been found further downstream at Green Road or Blackberry Way.



Time series plots showing how ARMI scores change over time across Birmingham River Champions sites along the upper Cole



Time series plots showing how ARMI scores change over time across Birmingham River Champions sites along the upper Cole tributaries

Across the Cole sites and their tributaries, ARMI scores were often higher during spring (March-April) and autumn (September-October). This likely reflects the life cycles of mayflies and caddisflies, where many species mature up to late spring/early summer, then emerge as winged adults before laying eggs that form into new generations that start becoming abundant again in autumn.

Lower scores were more common during summer (light grey) and winter (dark grey), which was clear for the Chinn Brook site in particular. This could be down to life cycles, but also could be when pollution is intensified by low river flows (/dilution) or high runoff washing contaminants into the channel during wet weather.

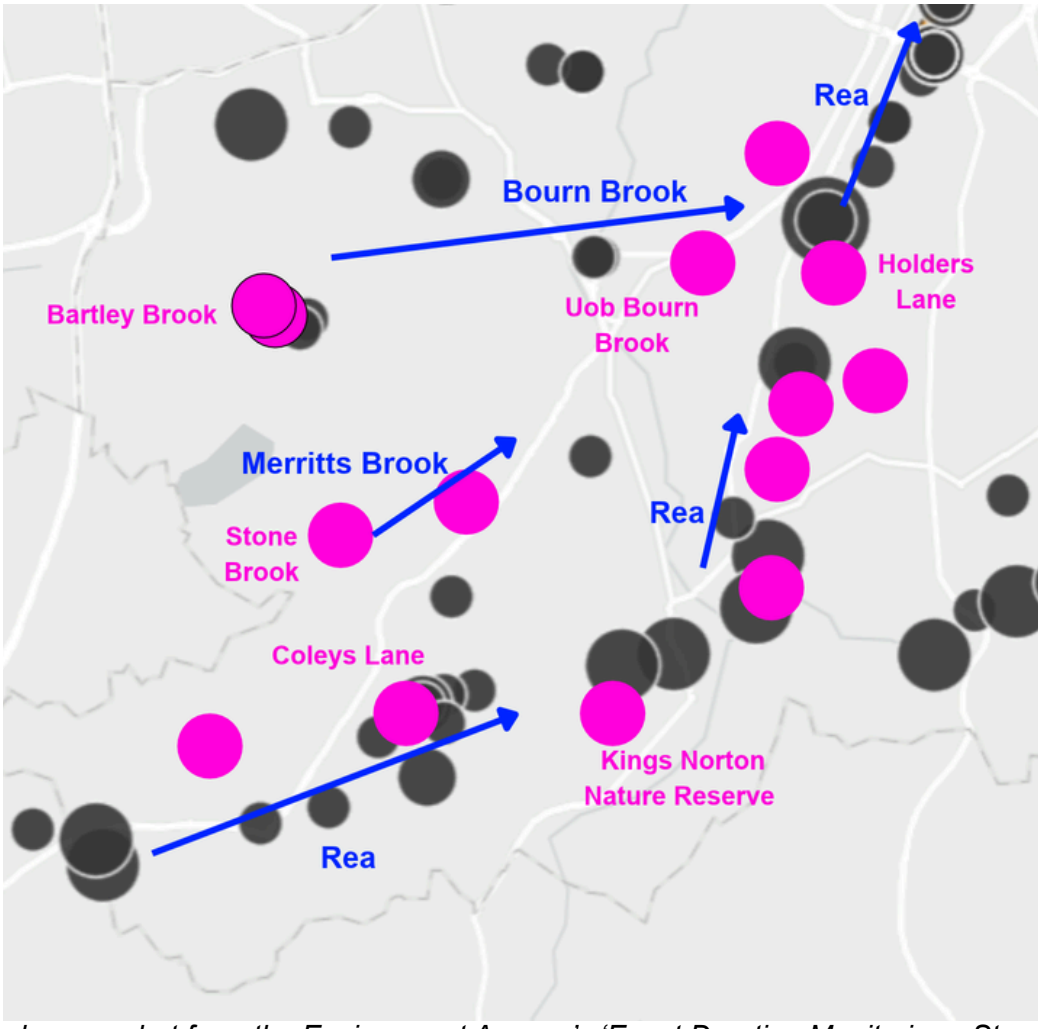
Notable declines occurred across all four Cole sites and Coldbath Brook during summer 2024 (but surprisingly not the Chinn Brook). Several low-scoring samples were collected at Scribers Lane during summer 2025, which volunteers suggested these declines may have been partly due to difficulties sampling in low-flows – although low ARMI scores occurred (albeit not as consistently) at all three tributary sites.

Only the Cole at Scribers Lane and Haunch Brook have displayed ARMI scores declining to the trigger level threshold, which occurred in just a single sample during last winter – Environment Agency can investigate if trigger levels are breached twice consecutively.

The Rea

The following sites along Rea and its tributaries have had more frequency sampling (a benchmark of six sites has been used): Coleys Lane, Kings Norton Nature Reserve and Holders Lane in Cannon Hill Park for the main Rea, and Stone Brook in Ley Hill Park, Bartley Brook in Senneleys Park and Bourn Brook on university campus.

The Rea has a greater number of CSOs compared to the Cole and extends to its headwaters near Rubery, and these pressures are also exposed to the Bourn brook (the university campus site); the remaining two tributaries are not subjected to these sewage inputs, but likely are exposed to other urban pressures described above.



A annotated screenshot from the Environment Agency's 'Event Duration Monitoring - Storm Overflow Explorer' (see link above). The black circles indicate CSOs, with larger sizes reflecting longer periods of time sewage is being released. The pink circles denote Birmingham River Champions sampling sites.

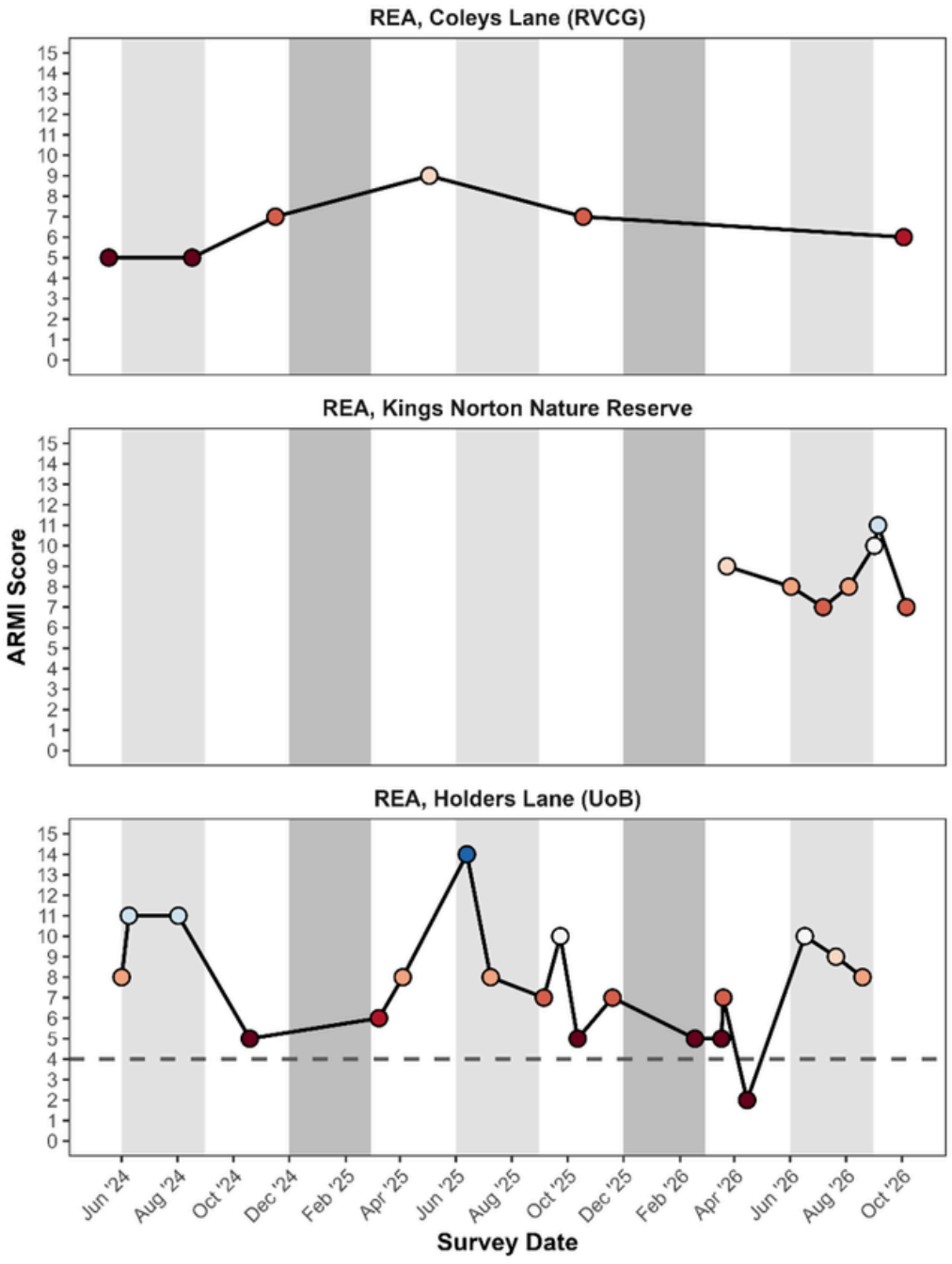
Comparing ARMI time series along the main Rea is more challenging than the Cole due to differences in sampling dates, and also these sites are more spread out.

However, opposite to what we found in the Cole, the Rea in Cannon Hill Park appears to be display lower ARMI scores during spring and autumn months, notably dropping below its trigger level in April this year (but only once).

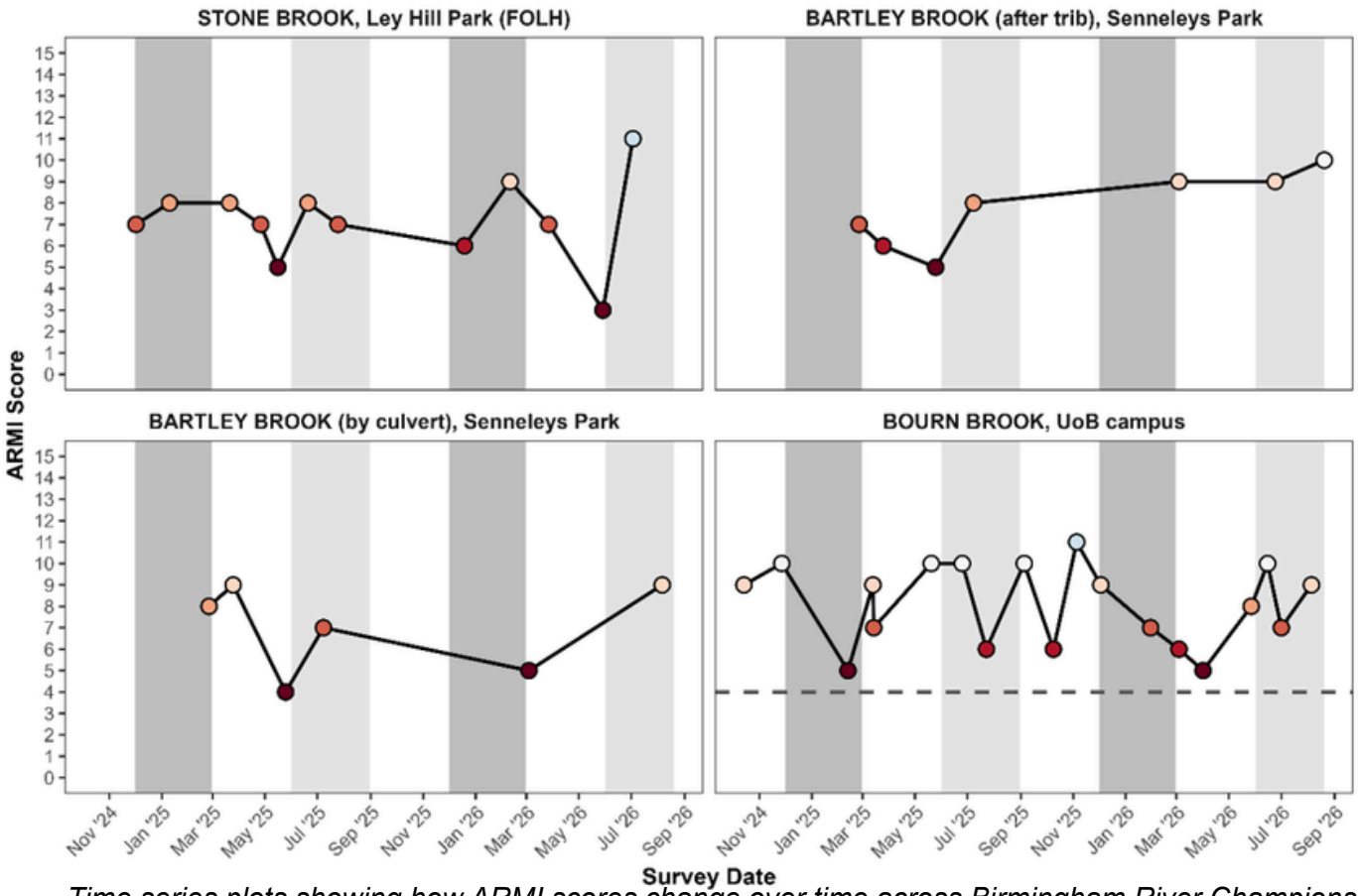
This was also observed at the Stone Brook and Bourn Brook – although the latter never declined to its trigger level set by the Environment Agency. The more recent sampling at Kings Norton Nature Reserve seems to be unveiling more predictable patterns where declines in summer (when mayflies and caddisflies have emerged and therefore not sampled in the water) were recorded and ARMI scores have been recovering since.

The Rea at Cannon Hill park displayed considerable variability, with ARMI scores ranging between 2-14. Particularly low values were recorded in late winter to spring this year, but seem to have picked up slightly as the year has gone on. Encouragingly, strong increases in ARMI scores were observed at the Stone Brook in Ley Hill Park recently, and to a lesser extent both sites along the Bartley Brook in Senneleys Park.

It should also be noted that a verified sampled stonefly was found in the Bartley Brook earlier in the year, which would suggest that it can (or has) provided suitable water quality conditions for sensitive species – and also indicates that the ARMI score may not be responsive enough to such indicators.



Time series plots showing how ARMI scores change over time across Birmingham River Champions sites along the Rea



Time series plots showing how ARMI scores change over time across Birmingham River Champions sites along tributaries of the Rea

The Tame Tributaries

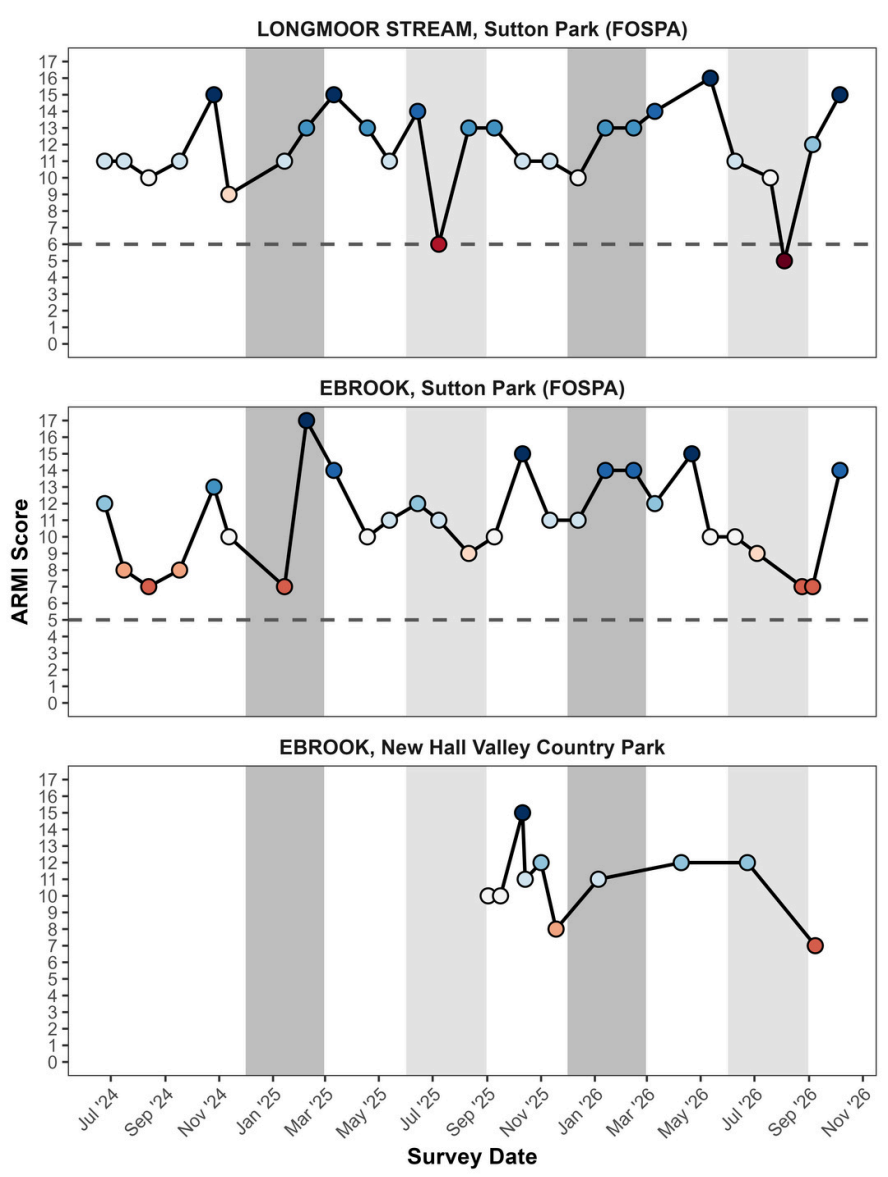
While BRC are yet to have a sampling site on the main watercourse, its Longmoor Stream and Ebrook tributaries in Sutton Park have been regularly sampled, and more recently so has the downstream Ebrook site at New Hall Valley Country Park.

These streams have displayed the highest ARMI scores observed by BRC, which reflects the presence of pollution-sensitive species described above. These streams also benefit from being located within large areas of semi-natural greenspace and, compared with many other monitoring sites across the city, are subjected to only two CSOs that discharge relatively infrequently – but as emphasised above, other pressures and sewage influence can still arise.

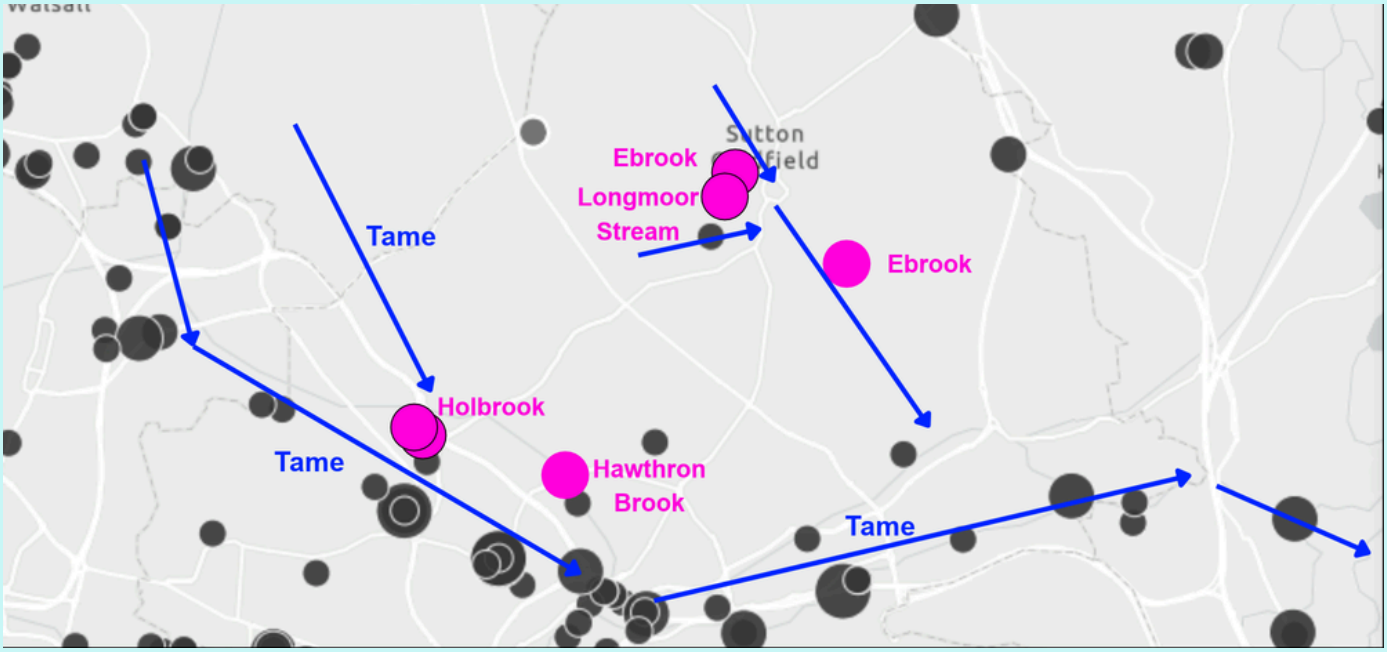
As found for the Cole, higher ARMI scores were generally recorded during spring and autumn. This again likely reflects the life cycles of aquatic insects, which are particularly abundant here, whereby the number of their larvae increase in the build up to and after they have emerged as winged adults in summer.

Although ARMI scores in the upper Tame tributaries were generally high, some declines were observed during summer months (see right). This was most clear in the Longmoor Stream, where ARMI scores dropped to, or below, the trigger level during the summers of 2024 and 2025, but this only happened once and recovered after.

It is plausible that summer declines in Longmoor Stream could be associated with an upstream CSO (see below) that can discharge at this time, but this wouldn't be clear without more detailed investigations.



Time series plots showing how ARMI scores change over time across Birmingham River Champions sites along the Tame.



A annotated screenshot from the Environment Agency's 'Event Duration Monitoring - Storm Overflow Explorer' (see link above). The black circles indicate CSOs, with larger sizes reflecting longer periods of time sewage is being released. The pink circles denote Birmingham River Champions sampling sites.

To Conclude...

Overall, the Urban Riverfly citizen science data collected has afforded invaluable insights into how the ecological health of Birmingham's rivers can vary in space and time. Sutton Park evidently supports the most biodiverse rivers regionally (of those sampled), but some pollution sensitive species have been observed in areas of the upper Cole and Bartley Brook at Senneleys Park.

Critically, the longer-term datasets haven't shown any strong evidence of consistent declines in ARMI scores, with most sites tending to show no obvious trend over time but there is strong variability within this. Moreover, of those sites that have had trigger levels set by the Environment Agency, none have breached this twice consecutively. Spring and autumn are typically the times to see the highest biodiversity, which is to be expected, although this seasonal trend was less clear on the Rea where fewer monthly datasets exist. Summer and winter months can lead to depressed ecological values.

We will continue trying to interrogate this data to see what this can tell us about the health of Birmingham's rivers and how this can be improved. This includes looking into the calculation of the ARMI metric and if this can be improved; more closely examining how specific species are responding to add a bit more meaning to the ARMI changes; and assessing how fluctuations in ARMI scores match up with water chemistry observations and other pollution indicators like CSO spills. If you have any thoughts or questions on this, please do not hesitate to contact us via the details below!

Birmingham River Champions

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